

# **MODULE-1**

## **Session 1**

### **(TASKS)**

#### **1. Name 5 IT job roles and write one line about what each role does.**

**Ans:**

- **Data Scientist** – Analyses large amounts of data to find useful insights and support decision-making.
- **Cloud Engineer** – Designs, deploys, and manages cloud-based infrastructure and services.
- **AI Engineer** – Develops and implements artificial intelligence and machine learning solutions.
- **System Analyst** – Studies business requirements and helps design IT systems to meet organizational needs.
- **IT Support Specialist** – Provides technical assistance and resolves hardware, software, and network issues for users.

#### **2. List 3 product companies and 3 service companies with one example of work each does.**

**Ans:**

##### **3 Product Companies and Their Work: -**

- **Google** – Develops products such as Android, Chrome, and Google Workspace.
- **Intel** – Manufactures computer processors and related hardware products.
- **Dell Technologies** – Produces laptops, desktops, servers, and storage devices.

##### **3 Service Companies and Their Work: -**

- **Accenture** – Provides IT consulting and business transformation services.
- **HCLTech** – Offers software development, infrastructure management, and cloud services.
- **Tech Mahindra** – Provides digital transformation, network, and software services to organizations.

#### **3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.**

**Ans:**

##### **Full Forms and One-Line Uses: -**

## Session 1 (TASKS)

- **GitHub** – Git Hub  
**Use:** A platform used to store, manage, and collaborate on software development projects using Git.
- **Slack** – Searchable Log of All Communication and Knowledge  
**Use:** A team communication and collaboration tool used for messaging and file sharing.
- **Jira** – Not an acronym (it is a product name)  
**Use:** A project management and issue-tracking tool used to plan, track, and manage software projects.
- **IDE** – Integrated Development Environment  
**Use:** Software that provides tools for writing, editing, compiling, and debugging code.
- **API** – Application Programming Interface  
**Use:** A set of rules that allows different software applications to communicate with each other.

### 4. Name any 5 programming languages and mention one popular use case for each.

**Ans:**

#### 5 Programming Languages and Their Popular Use Cases: -

- **Python: -**  
**Use Case:** Artificial Intelligence (AI) and Machine Learning: -  
  
Python is widely used for AI, machine learning, data science, and automation because of its simple syntax and powerful libraries such as TensorFlow, PyTorch, and NumPy.
- **Java: -**  
**Use Case:** Android Application Development: -  
  
Java is a popular programming language used for developing Android mobile applications. It is also widely used in enterprise-level applications due to its platform independence and reliability.
- **C++: -**  
**Use Case:** Game Development and System Software: -  
  
C++ is used for developing high-performance applications such as video games, operating systems, and embedded systems because it provides fast execution and efficient memory management.
- **JavaScript: -**  
**Use Case:** Interactive Web Development: -

## Session 1 (TASKS)

JavaScript is used to create dynamic and interactive websites. It enables features such as animations, form validation, real-time updates, and interactive user interfaces in web browsers.

- **PHP: -**

**Use Case: Server-Side Web Development: -**

PHP is commonly used to build dynamic websites and web applications. Popular platforms such as WordPress use PHP to manage content, databases, and user interactions on websites.

### 5. What is the difference between a developer and a tester? Answer in 2 lines.

**Ans:**

- **Developer:** A developer is responsible for designing, writing, and maintaining the source code of a software application. They create new features, fix issues, and ensure that the software functions according to the requirements.
- **Tester:** A tester is responsible for examining the software to identify bugs, errors, and performance problems. They test different functionalities of the application and verify that it works correctly, reliably, and meets quality standards before it is released to users.